Ayan Majumdar

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About

- I am a Doctoral Researcher at MPI-SWS specializing in Trustworthy AI. I am passionate about bridging the gap between theoretical research in fairness, causality, and explainability and the practical deployment of safety-compliant Automated Decision-Making and Generative AI systems.

Research Interests

Trustworthy AI	Responsible AI, AI Safety, Fairness, Explainability, Causality
Deep Learning	Generative AI, Large Language Models, Vision Language Models, Model Alignment

Education

2021 – 2025 (expected)

Ph.D. in Computer Science
MPI-SWS and Saarland University, Saarbrücken, Germany
Area: *Deep Learning, Generative AI, Fairness, Explainability*
Advisor: Prof. Dr. Krishna P. Gummadi, Prof. Dr. Isabel Valera
Contact: gummadi[at]mpi-sws.org, ivalera[at]cs.uni-saarland.de

2017 – 2021

M.Sc. in Computer Science
Saarland University, Saarbrücken, Germany
Thesis: *Generating Counterfactuals for Causal Fairness*
Outline: *Deep generative models and their implicit assumptions in generating counterfactuals from observed data in the context of fairness.*
Advisor: Prof. Dr. Krishna P. Gummadi, Prof. Dr. Isabel Valera
GPA: 1.2/1.0 (German Scale)

2011 – 2015

B.Tech. in Electronics & Communication
Heritage Institute of Technology, Kolkata, India
Project: *Automated traffic detection using image processing*
Outline: *Utilize blob detection techniques for detecting traffic from video sequences.*
Supervisor: Prof. Anindya Sen
GPA: 8.8/10.0

Publications

Conference Proceedings

1. **Majumdar, A.**, & Valera, I. (2024). CARMA: A practical framework to generate recommendations for causal algorithmic recourse at scale. In *2024 ACM Conference on Fairness, Accountability, and Transparency* (pp. 1745–1762). Paper Code.
2. Nanda, V., **Majumdar, A.**, Kolling, C., Dickerson, J. P., Gummadi, K. P., Love, B. C., & Weller, A. (2023). Do invariances in deep neural networks align with human perception? In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 37, pp. 9277–9285). Paper Code.
3. Rateike*, M., **Majumdar*, A.**, Mineeva, O., Gummadi, K. P., & Valera, I. (2022). Dont Throw it Away! The Utility of Unlabeled Data in Fair Decision Making. In *2022 ACM Conference on Fairness, Accountability, and Transparency* (pp. 1421–1433). Paper Code.

Archival Preprints

4. **Majumdar*, A.**, Kanubala*, D. D., Gupta, K., & Valera, I. (2026). A causal framework to measure and mitigate non-binary treatment discrimination. *arXiv preprint arXiv:2503.22454*. Accepted to AAAI 2026 Paper Code.
5. **Majumdar, A.**, Chen, F., Li, J., & Wang, X. (2025). Evaluating llms for demographic-targeted social bias detection: A comprehensive benchmark study. *arXiv preprint arXiv:2510.04641*. Paper.

Work Experience

Aug. 2024 – Feb. 2025

Ph.D. Research Intern
Huawei Research TTE-DE Lab, München, Germany
Project: Scaling-up Data Bias Detection & Survey on AI Governance
Role: Developed a benchmark using LLMs to detect demographic-targeted social bias in large datasets (preprint [5]). Conducted research surveys to support AI governance and alignment with regulations like the EU AI Act.

Oct. 2019 – Mar. 2021

Research Assistant
Max Planck Institute for Software Systems, Saarbrücken, Germany
Project: *Exploring bias and fairness with deep generative models*
Role: Led project regarding exploration of bias in deep generative models for facial image data; designed core methodologies and evaluations.
Supervisor: Prof. Dr. Krishna P. Gummadi

Apr. 2018 – Mar. 2019

Research Assistant
SFB1102, Saarland University, Saarbrücken, Germany
Project: *Mutual Intelligibility in Slavic Languages*
Role: Engineered automated data processing pipelines to collect and clean large-scale textual datasets for Machine Translation experiments. Developed web-based user studies to evaluate mutual intelligibility in Slavic languages.
Supervisor: Prof. Dr. Dietrich Klakow

Jul. 2015 – Aug. 2017

Systems Engineer
Infosys Ltd., Bengaluru, India
Role: Optimized and validated Session Initiation Protocol (SIP) and VoIP functionalities within session border controllers. Recognized with the Infy Insta Award (2017) and Spot Award (2016) for critical project contributions

Technical Skills

Generative AI & LLM Engineering

Frameworks: Transformers, vLLM, Sentence-Transformers, FAISS
Techniques: Prompt Optimization, In-Context Learning, Multi-modal Inference, RAG, Fine-tuning, Benchmarking, LLM-as-a-Judge, Automated Content Moderation

Responsible AI & AI Safety

Frameworks: AIF360, Fairlearn, CleverHans, Foolbox, robustness
Techniques: Causal Inference, Counterfactual Explanations, Statistical & Causal Fairness

Technical Skills (continued)

Programming & Core Infrastructure

Core ML	PyTorch, Lightning, Scikit-learn, NumPy, Pandas, Scipy, Optuna
Languages & Tools	Python, Git, LaTeX, Weights & Biases, Matplotlib, Seaborn

Teaching Assistance

Summer 2024	**Machine Learning (Core Lecture)**, Saarland University
Summer 2021	**Machine-Assisted Decision Making (Seminar)**, Saarland University
Summer 2019	**Statistical Natural Language Processing (Advanced Lecture)**, Saarland University

Mentoring Experience

May 2024 – July 2024	Co-mentored Pranav Nyati (B.Tech. student, IIT-Kharagpur) during DAAD summer internship on **algorithmic recourse using swarm algorithms**.
Sept. 2025 – Jan. 2026	Co-mentored Lena Marie Budde (M.Sc. student, Saarland University) during thesis and project on **personalization in algorithmic recourse**.

Talks and Posters

2024 — **Scaling Up Causal Algorithmic Recourse with CARMA**
European Workshop on Algorithmic Fairness 2024, Center for Perspicuous Computing 2024 Meeting

2022 — **Don't Throw it Away! The Utility of Unlabeled Data in Fair Decision Making**
SAP Innovation Sessions, MPI for Intelligent Systems, MILA Quebec AI Institute

Online Certifications

Coursera	Algorithmic toolbox, Data structures, Graph algorithms, String algorithms

Academic Activities

Program Committee	FAccT 2024-25, ICLR 2023-25, ICML 2023-24, NeurIPS 2023/25, WebConf-Web4Good 2026, AAAI 2025-26, AIES 2025

Language Skills

Native/Fluent	English	●●●●●●	Bengali	●●●●●●
Proficient/Intermediate	German	●●●○○○	Hindi	●●●○○○

Achievements & Awards

- Infy Insta award (2017) and Spot award (2016) for significant contributions to the project, Infosys, India.